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Multi-Target Threat Assessment for Automotive Applications

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Abstract— **Auto-brake systems have been on the market for a few years, but in order to continue improving their performance, multi-target threat assessment is a key component. The goal is to early be able to say that a collision can no longer be avoided by a steering maneuver, and thus braking can be started earlier. This is done by looking at all objects in the road scene, including other vehicles and barriers, and then looking for escape paths. The paper presents a computationally efficient way of determining whether an escape path can be found in a scenario with generally positioned objects, given the limited dynamics of the vehicle.**

I. INTRODUCTION

More and more people agree that auto-brake will be the next big step in automotive crash mitigation and prevention. By automatically braking the vehicle just before an imminent collision, the collision speed can be reduced and in some cases, the accident can be avoided completely. The Swedish research institute Folksam states that if the impact velocity in collisions in general where reduced by on average 20 km/h, the risk of being seriously injured would be reduced by 40% [1].

Auto-brake systems have now been on the market for a few years, and current state of art is an autonomous speed reduction of maximally around 35 - 40 km/h [2].

Typically, auto-brake systems are based on one or more sensors that monitor the forward zone of the vehicle. The sensors may for instance be radars [3], laser scanners, mono-vision or stereo-vision [4], or a combination of these [2]. The sensors feed information about surrounding objects to a decision module that decides what kind of preventive action that should be taken. The most common approach is to provide a visual and acoustic warning two to three seconds before impact, and then if the driver does not brake - or does not brake enough - the vehicle will either help the driver brake or provide autonomous braking at some level.



Fig. 1. A vehicle approaching a stationary or slower moving vehicle. H stands for Host and indicates which vehicle is equipped with the auto-brake system.

It would be natural to think that the development will continue and that the next generation of auto-brake systems would reduce the velocity even more. However, 40 km/h of speed reduction seem to be a fundamental limit, and it is very difficult reduce the impact velocity more. The reason is that the systems are unaware of the drivers intentions.

When the driver is approaching an obstacle, say a stationary leading vehicle, the driver has the options of steering or braking to avoid an accident (see Fig. 1). At low speeds, below approximately 40 km/h, the braking option is much more effective, i.e., braking can be commenced later than steering and the accident can still be avoided. At higher speeds, however, steering is more effective, i.e., braking has to be commenced earlier than steering in order to avoid the obstacle. Braking cannot be commenced until it is judged that the obstacle can no longer be avoided by steering, since the driver might be planning to steer, for instance to do a late lane change. This is in order to minimize interventions that are perceived as incorrect or disturbing. At this point it is too late to avoid the collision by braking, so the only remaining option for the safety system is to try to reduce the impact velocity.

Furthermore, in order to avoid an accident at a relative speed of around 40 km/h, braking has to be commenced at or just before 1 s Time-To-Collision (TTC). Since the relative velocity changes during the maneuver, the time that the system has to brake the vehicle increase, maximally to double the initial TTC, in this case a little over 2 s. However, this effect decreases with increasing relative velocity, i.e., the braking duration is reduced and approaches the initial TTC value, in this case 1 s. This means that the potential speed reduction also decrease. In fact, the potential speed reduction at higher speeds will approach a value less than half of the maximal speed reduction for lower speeds. The exact relation between the two depends on the brake system and on how quickly deceleration can be generated.

The maximal speed reduction can be somewhat improved through the development of more powerful brake systems, but probably not to a level much higher than 40 km/h.

II. MULTI-TARGET

This is where the concept of multi-target comes in. Consider the same example again, the driver is approaching a stationary or slower moving vehicle, but this time the steering option is blocked, for instance by another vehicle or by a barrier, see Fig. 2. In this case the driver cannot avoid the collision by steering, and thus, if the auto-brake system in the host vehicle could take these objects into account, braking could be commenced much earlier.

Similarly, Fig. 3 show an example where the host is approaching a slower moving vehicle, the path to the left is

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blocked by a barrier and the path to the right is blocked by a faster vehicle in the adjacent lane. Again, if these objects could be considered by the auto-brake system, then braking could be commenced earlier.

Very little research has been carried out in this field, in particular for automotive applications. The topic has been addressed in naval applications [5] and in robotics [6], although with very different prerequisites. In these applications, all the participants can often be controlled, and the goal is to get them to a predetermined position. This is very different from the present application, where we can only control one object and want to know if a solution can be found, given the limited dynamics of the vehicle.

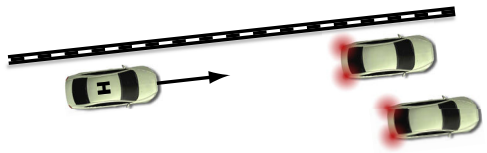


Fig. 2. A vehicle approaching a stationary or slower moving vehicle again. This time the steering option is blocked by a barrier and another stationary vehicle in the adjacent lane.

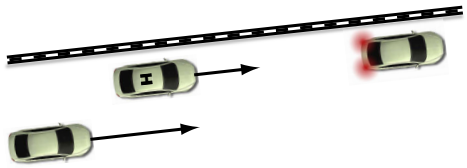


Fig. 3. A vehicle approaching another stationary. This time the steering option is blocked by a barrier and faster moving vehicle approaching from behind in the adjacent lane.

III. THREAT ASSESSMENT

A. Single target

As was stated in the introduction, the goal is to minimize the number incorrect or disturbing interventions. That is why an intervention only can be started when a collision is unavoidable or close to unavoidable.

One way to achieve this is to compute something called Brake Threat Number (BTN) and Steering Threat Number (STN) [7]. BTN is defined as the required deceleration level to avoid the collision ($a_{\text{long,req}}$) divided by the maximally available deceleration that the vehicle can achieve ($a_{\text{long,max}}$), which is limited by tire-to-road friction and brake performance. Similarly, STN is defined as the required lateral acceleration required to avoid the obstacle ($a_{\text{lat,req}}$) divided by the available lateral acceleration ($a_{\text{lat,max}}$), which is also limited by tire-to-road friction, but also the design of the steering system in the vehicle, and the drivers ability to turn the steering wheel. Formally,

$$\text{BTN} = \frac{a_{\text{long,req}}}{a_{\text{long,max}}} \quad \text{and} \quad \text{STN} = \frac{a_{\text{lat,req}}}{a_{\text{lat,max}}} \tag{1}$$

These tell you how much of the available vehicle capability has to be used to avoid an accident. When these equal 1, you know that the collision is unavoidable, and the vehicle auto-brake system may commence an intervention. This may sound very conservative, but it is very efficient if you want to avoid interfering with normal driving. If a less conservative system is desired, then the available acceleration can instead reflect what the driver is expected to use, rather than what the vehicle is capable of. If drivers never steer more than a certain amount, then this amount could be used as $a_{\text{lat,max}}$. This would trigger earlier interventions.

For a situation like the one in Fig. 1 BTN and STN are easy to compute, depending on how accurate physical model of the vehicle that is needed. The model can for instance be constant acceleration where it is assumed that the vehicle instantly can achieve a lateral and longitudinal acceleration, and that the lateral and longitudinal motion are independent [7]. More advanced models have also been proposed, for instance constant longitudinal jerk and constant steering wheel angular velocity [8]. If such a model is used, BTN and STN could be computed based on other properties as well. It could for instance be the required steering wheel angular velocity divided by the maximal angular velocity. However, this article will mainly focus on the (piecewise) constant acceleration model.

In Fig. 1, computing BTN and STN is straight forward, in particular if a constant acceleration model is used. The required longitudinal acceleration $a_{\text{long,req}}$ is just the acceleration required to bring the host vehicle to range-rate zero at impact. The required lateral acceleration $a_{\text{lat,req}}$ is the smallest of the accelerations required to avoid the obstacle to the left or right.

B. Multi target

In Fig. 2, computing BTN and STN is a little more complicated, but a human observer can immediately see that the required lateral acceleration corresponds to passing to the right of the two slower moving vehicles. It gets a little more tricky in Fig. 3. Here the vehicle has to either steer sharp right in order to clear the vehicle in the adjacent lane before it reaches the host vehicle, or wait a little bit and then steer once it has passed. The required lateral acceleration is the smallest of those two options.

The question is how to write an algorithm for this. The sensors typically generate a list of objects with positions, velocities etc. for all objects. From that list the algorithm has to figure out which maneuvers that clears all the objects. Furthermore, among these maneuvers, we are interested in the "optimal" one in the sense that it generates the least lateral acceleration. This is really how the required lateral acceleration is defined.

This can be formulated mathematically:

$$\min_{m \in \mathcal{M}} C_m(m) \tag{2}$$

where m is a real function $m(t)$ which defines the maneuver, $\mathcal{M}$ is the set of *conflict free* maneuvers and C_m is the "cost" function. The function $m(t)$ may be equal to the lateral

acceleration along the path, i.e., $m(t) = a_{\text{lat}}(t)$, but may also be something else that defines the path, for instance steering wheel angle. The function C_m defines the cost for a maneuver, here this will be equal to the maximal lateral acceleration along the path such that $C_m(m) = C_m(a_{\text{lat}}) = ||a_{\text{lat}}||_\infty = \max_t |a_{\text{lat}}(t)|$. In that case (2) becomes

$$a_{\text{lat,req}} = \min_{a_{\text{lat}} \in \mathcal{M}} \max_t |a_{\text{lat}}(t)| \tag{3}$$

The problem is that the search space $\mathcal{M}$ is high dimensional since the maneuver changes along the way. It is also non-convex, which can be seen by the simple example in Fig. 1. The object can be passed either to the left or to the right. The shape of $\mathcal{M}$ may also be very complex since it is defined by the position and shape of other objects, which are also typically moving.

A straight forward approach would be to discretize the time and then apply conventional optimization methods in order to find the optimal solution. However, since the problem is high dimensional and non-convex it will be computationally very costly. If the time is discretized into 0.1 s timesteps and a three second time horizong is used, then the optimization space will have 30 dimensions. Remember that this has to be executed in real time in an on-board control unit. In order to overcome this problem, three different methods have been proposed [7].

- *Constant Acceleration* Restrict the search space to constant maneuvers only.
- *Heuristic Control* Restrict the search space to specific maneuvers, for instance go straight a certain time and then steer.
- *Exhaustive Search* Discretize time and the lateral acceleration space and search for avoidance maneuvers. This is very computationally expensive since, if the number of time steps is N and the number of acceleration levels being considered is K, then the total number of maneuvers is K^N.

For more details about these methods, readers are referred to the original source. In all three cases, only a small subset of the conflict free paths $\mathcal{M}$ is or can be considered.

It should be clarified to the reader that the goal is not to steer the host vehicle along the optimal path. The goal is to judge if it is possible to avoid a collision through a steering maneuver. If the lateral acceleration required to follow the optimal path is higher than what the vehicle is capable of, we know that the collision cannot be avoided by a steering maneuver. In that case auto-braking can be commenced without any risk of interfering with the drivers intentions.

IV. PROPOSED ALGORITHM

A. Overview

The proposed solution uses the assumption that the optimal maneuver will always be of cost zero, i.e., going straight forward, or touching at least one of the objects. The idea is similar to what is used in linear programming. Either the optimum is zero, or that point is blocked by one or more of the constraints. In that case the optimum will lie on the constraint, as close to zero as possible.

Since the algorithm does not know which of the constraints (the obstacles) that will be the "active" ones, the idea is to create one maneuver for each constraint. An object can be passed either to the left or to the right, which in general means two maneuvers per object. Moreover, since we would like to include maneuvers that change along the way, we also need to include paths that first touch one object and then another. This means including all combinations of maneuvers also, e.g., passing to the left of object i and then to the right of object j.

This can be implemented recursively in a natural way. First compute the maneuvers that will pass all objects to the left and to the right, see Fig. 4. Then, for each such maneuver, predict the future positions of all objects in the road scene, given that the host follows that maneuver. Given these new positions, call the algorithm again in order to compute the remainder of the maneuver, see Fig. 5. Each call to the algorithm will return the maneuver with the lowest cost. This cost is computed as the maximum of the cost of the current segment and any of the future segments.

The result is a kind of search tree, with each branch ending when either all objects are cleared, or when no escape path is found, i.e., when the path is blocked.

B. Definitions

x_i^t	State vector for object (other vehicles, pedestrians etc.) i at time t.
y_i^t	Barrier state vector at time t.
z^t	Host vehicle state vector at time t.
TTC_i	Time to collision for object i
$p = (m, t)$	A maneuver/time pair of the host vehicle where m is a maneuver of the host vehicle and t is the length in time of this maneuver.
C_m	Maneuver cost function; $C_m(p)$ is the cost of maneuver/time pair p. The cost may be independent of the time and in that case, if $p = (m, t)$, we have $C_m(p) = C_m(m)$.
P_{obj}	Prediction model for objects: $x_i^{t+\Delta t} = P_{\text{obj}}(x_i^t, m, \Delta t)$ where m is a host vehicle maneuver and Δt is the prediction time.
P_{barrier}	Prediction model for barriers: $y_i^{t+\Delta t} = P_{\text{barrier}}(y_i^t, m, \Delta t)$. The barriers may or may not move depending on the choice of coordinate system. If they do not move, then $y_i^{t+\Delta t} = y_i^t$ for all Δt.
P_{host}	Prediction model for the host vehicle. $z^{t+\Delta t} = P_{\text{host}}(z^t, m, \Delta t)$.

C. Algorithm

The algorithm starts at time zero with a set of objects with state vectors x_i^0, a set of barriers with state vectors y_i^0 and a host vehicle state vector z^0. These are the positions that are given by the sensors. During the computation of the optimal maneuver, object positions are updated, but these predictions

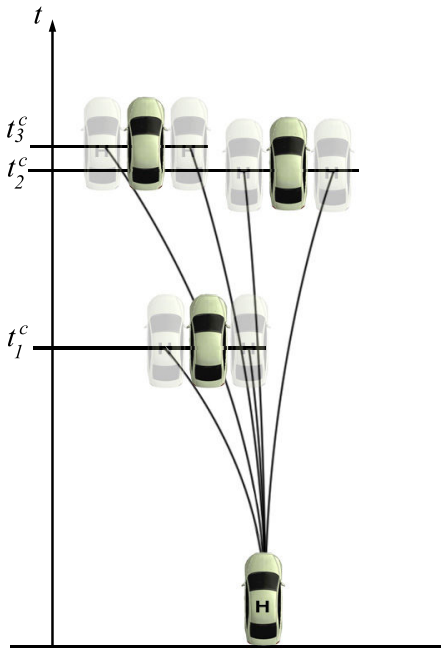


Fig. 4. In the first step of the algorithm, the maneuvers to pass each object just to the left or right are computed.

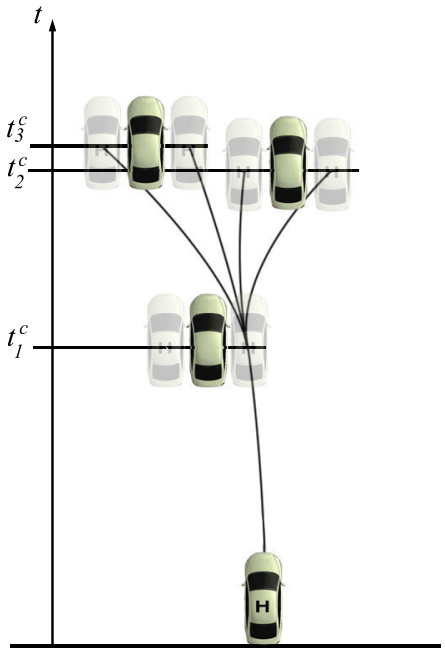


Fig. 5. The second step of the algorithm. For each maneuver in step 1 (See Fig. 4 maneuvers to pass the remaining objects are considered.

are purely virtual. No objects are actually moved in the way the algorithm suggests.

1) If there are no objects or barriers, then abort the algorithm and return the cost zero.
2) For each object i, compute the maneuver/time pairs $p_i^R = (m_i^R, \text{TTC}_i)$ and $p_i^L = (m_i^L, \text{TTC}_i)$. Here, m_i^R and m_i^L are the cheapest maneuvers (with the lowest cost according to C_m) required to tangent object i on the left and right side, respectively, and TTC_i is the time to collision for object i. The maneuvers m_i^R and m_i^L are only valid until the time TTC_i, see the Fig. 4. Form the sets $M^R = \{p_i^R, i = 1\ldots\}$ and $M^L = \{p_i^L, i = 1\ldots\}$.
3) For each barrier j which is on collision course, compute the maneuver/time pairs $p_j^B = (m_j^B, \text{TTC}_j)$. Here, m_j^B is the cheapest maneuver required to tangent barrier j and TTC_j is the time to collision for barrier j given the maneuver m_j^B. Form the set $M^B = \{p_i^B, i = 1\ldots\}$.
4) Form the common set of maneuvers $M = M^R \cup M^L \cup M^B$.
5) Add a maneuver/time pair $p_0 = (m_0, t_0)$ to the set M where m_0 is the cheapest possible maneuver not considering any objects or barriers and $t_0 = \infty$. This typically means that the host vehicle keeps going straight and the cost of this maneuver is typically zero.
6) Remove all conflicting maneuver/time pairs, i.e., if a pair $(m, t) \in M$ lead to a conflict with an object i with $\text{TTC}_i < t$ or any of the barriers, then remove the pair from M.
7) If the maneuver p_0 is still in the set M, i.e., if p_0 is conflict free, then abort the algorithm and return the cost $C_m(p_0)$. Typically $C_m(p_0) = 0$.
8) If the M is empty, then there are no escape paths, return the cost ∞.
9) For all remaining maneuver/time pairs $p_k = (m_k, t_k)$ in the set M, repeat:
   a) Predict the state of all objects at time t_k, i.e., compute $x_i^{t_k} = P_{\text{obj}}(x_i^0, m_k, t_k)$.
   b) Predict the state of all barriers at time t_k, i.e., compute $y_j^{t_k} = P_{\text{barrier}}(y_j^0, m_k, t_k)$.
   c) Predict the host vehicle state at time t_k, i.e., compute $z^{t_k} = P_{\text{barrier}}(z^0, m_k, t_k)$.
   d) Remove all objects with $\text{TTC} < t_k$.
   e) Call the algorithm again with the reduced set of objects and all predicted states $x_i^{t_k}$, $y_j^{t_k}$ and z^{t_k}. For each maneuver, the algorithm will generate new maneuvers for the second step in the sequence, see Fig. 5. The algorithm returns a cost, say C_k^{pred}, associated with each predicted scenario.
   f) Form $C_k = \max(C_m(p_k), C_k^{\text{pred}})$. The final output of the algorithm is a sequence of maneuvers and by taking the max here the cost of the sequence will be the maximum of the cost of any of the maneuvers in the sequence.
10) Return $\min_k(C_k)$.

The result is a piecewise constant acceleration maneuver, and the required lateral acceleration for the total scenario can be computed as the maximum acceleration of all the segments. The algorithm is extremely computationally efficient, since very few maneuvers actually has to be computed. In particular when combined with a simple motion model like constant acceleration, then each maneuver can be computed with a simple analytical expression.

D. Computational cost

It can be shown that if the number of obstacles is n, then the number of maneuvers that has to be computed and considered is approximately $2n^3/3$. Each maneuver consists of computing a simple analytical expression. This means that in a scenario with 5 obstacles, a total of around 80 maneuvers has to be computed, something that can easily be carried out in real time. Even up to 10 obstacles should not be a problem in modern hardware. If more than 10 - 15 objects are present in the road scene, then some kind of pre-processing should probably be applied where non-relevant objects are discarded, e.g., objects with increasing distance or high TTC.

V. RESULTS

This section will show a number of simulated scenarios together with the output from the algorithm. The plots show other vehicles as rectangles and barriers are represented by dashed lines. In all scenarios, the initial position of the host is in the origin, traveling in the positive x-direction. The optimal maneuver is then plotted as two lines showing the motion of the left and right corners of the front of the host vehicle. The host can be imagined to travel in the corridor defined by the two curves. Although the algorithm can handle moving objects, all objects in these examples are stationary since it easier to illustrate. Fig. 6 shows the result of the scenario defined in in Fig. 1. The optimal path is to steer to the right of the obstacle. Fig. 7 shows the same scenario again, but with two additional vehicles blocking the path, which results in higher required lateral acceleration. Fig. 8 show a scenario with three vehicles, and it turns out that the optimal path is to pass between the first two objects and then steer to avoid the third object. The last two examples, Fig. 9 and Fig. 10, are situations that also include barriers. Situations like the one in Fig. 3 can also be evaluated, but the result is difficult to illustrate without animation.

The reader is again reminded that, here the intention is not to steer the host vehicle along this path, the goal is to estimate if it is possible to avoid the collision, although the same algorithm could be used for that purpose as well.

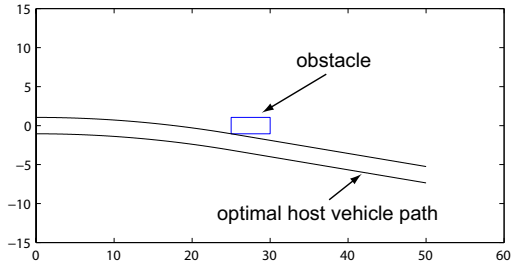


Fig. 6. A situation with a single obstacle which is illustrated by the rectangle. The x and y-axis are in meters. The initial position of the host is in the origin, traveling in the positive x-direction with a speed of 25 m/s = 90 km/h. This corresponds to the scenario defined in Fig. 1. The optimal maneuver of the host is then plotted as a corridor defined by the two curves. In this example, the optimal path is a lateral acceleration of -4.2 m/s^2 in 1.0 seconds and then going straight, and thus the required lateral acceleration is 4.2 m/s^2.

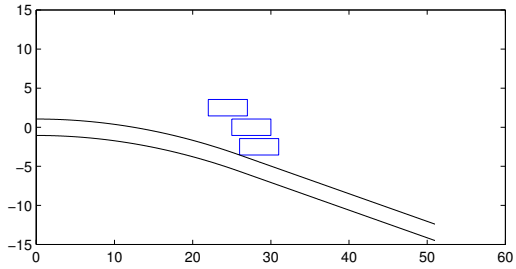


Fig. 7. The same scenario as in Fig. 6, but with two additional vehicles. The host speed is 25 m/s. The lateral acceleration required to avoid a collision increase to 8.5 m/s^2.

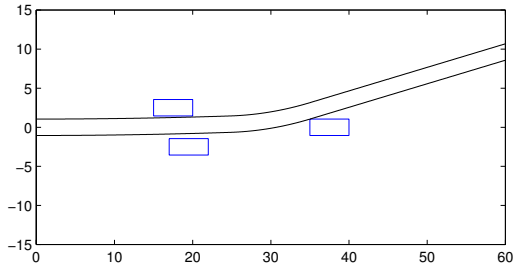


Fig. 8. In this scenario the optimal maneuver consists of two segments. First a lateral acceleration of 0.8 m/s^2 in 1.0 seconds followed by an acceleration of 16.5 m/s^2 in 0.4 seconds. After that all obstacles are cleared and the host can keep going straight ahead. This means that the required lateral acceleration for the scenario is max($|0.8|$,$|16.5|$) = 16.5 m/s^2. Again, the host vehicle speed is 25 m/s.

VI. DISCUSSION AND FUTURE WORK

A. Discussion

A method for computing the required lateral acceleration to avoid a collision in a general, multi-target scenario has been presented. The method is based on a recursive search-tree and uses the assumption that if the optimal path is not going straight forward, then it will always tangent at least one of the objects. By using this assumption the computational

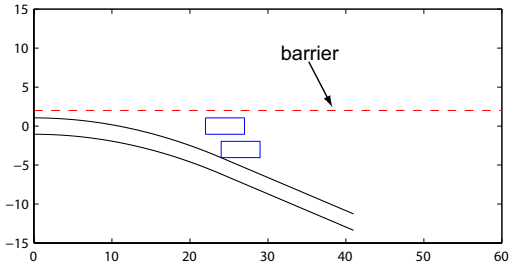


Fig. 9. This scenario also contains a barrier, represented by the dashed line, this corresponds to the scenario in Fig. 2 in the introduction. The host is traveling at 20 m/s = 72 km/h. Here, the optimal path is a lateral acceleration of -7.1 m/s^2 in 1.2 seconds.

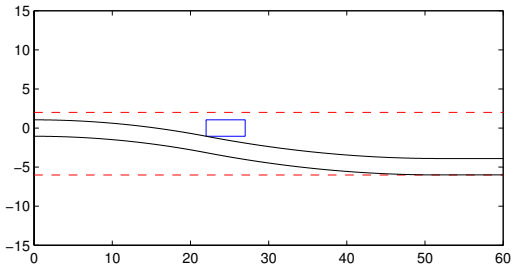


Fig. 10. This scenario has two barriers and a single obstacle. The host vehicle speed is again 20 m/s. The resulting maneuver consist of an acceleration of -3.5 m/s^2 in 1.1 seconds and then 2.6 m/s^2 in 1.5 seconds. Thus, the total required lateral acceleration is 3.5 m/s^2.

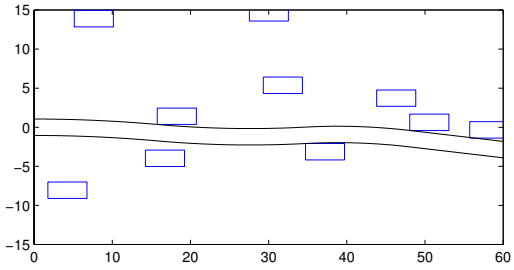


Fig. 11. Here, 10 obstacles was placed randomly just do demonstrate the power of the algorithm. The optimal path consist of three segments, with accelerations -1.3 m/s^2, 1.7 m/s^2 and -2.9 m/s^2, with the corresponding times 1.0 s, 1.3 s and 0.9 s.

cost of finding the optimal path can be reduced dramatically. The full problem means solving a non-linear and non-convex 30 dimensional optimization problem which is likely to be very difficult even off-line. The reduced problem is extremely computationally efficient and can easily be run in real time. This has also been demonstrated in a dSpace hardware at Volvo Cars.

It could be argued that this type of computation is very conservative, i.e., very few drivers would actually drive according to the trajectories found in the algorithm, and because of that, the brake intervention will start unnecessarily late. However, even if very few drivers do drive like that, triggering false interventions for those few very aggressive drivers is still unacceptable. Furthermore, if it turns out that drivers in general use more margin when passing an object, then a margin can be added to the size of the object when it is used in the algorithm, i.e., the optimal path means passing a certain small distance from the object. Also, if it can be shown that drivers never actually use more than a certain level of lateral acceleration, then the maximally available lateral acceleration can be set to that level when computing the STN value.

B. Future Work

This article is based on a simple motion model described by constant acceleration. It would be interesting to combine the optimization framework with a more advanced motion model, like the one in [8]. Initial testing in a vehicle has been carried out, but continuing these tests on a larger scale is of course also on the agenda.

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